# Nikita Kalinin

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clickable links:

Guangdong Technion profile

mathscinet profile  zbmath

ORCID  web of science  scopus

google scholar  arxiv

Interests: analytic/elementary number theory around $SL(2,\mathbb{Z})$, Farey fractions, topographs, and lattice sums; continued fractions, affine geometry.

## Positions

Associate professor (tenure track), *Guangdong Technion-Israel Institute of Technology*, Shantou, China (01.2024 – Present).

Visiting Associate professor, *Guangdong Technion-Israel Institute of Technology*, Shantou, China (10.2022 – 11.2023).

Associate professor (доцент), *Department of Mathematics and Computer Science*, Saint Petersburg University, Russia (09.2019 – 11.2022).

Senior Researcher (старший научный сотрудник), *International Laboratory of Game Theory and Decision Making, Saint Petersburg branch of HSE* (National Research University Higher School of Economics), Russia (09.2017 – 10.2022).

A holder of a Swiss grant PostDoc.Mobility with Ernesto Lupercio as a hosting professor, *CINVESTAV (Centro de Investigación y de Estudios Avanzados)*, Mexico City, Mexico (07.2016 – 08.2017).

## Education

2015 PhD in Mathematics, *University of Geneva*, Switzerland
Thesis topic: "Tropical geometry for Nagata's conjecture and Legendrian curves"
Advisor: Grigory Mikhalkin

2010 Specialist (=master) in Mathematics, *Saint Petersburg University*, Russia
Thesis topic: "The Alexander polynomial as an intersection of two cycles in a symmetric power"
Advisor: Oleg Ya. Viro

## Honors, Awards, and Grants

| | |
|---|---|
| 2026 | Technion-GTIIT seed grant, Probabilistic methods in geometry (joint with Ilya Gekhtman) |
| 2020 | Russian Science Foundation grant №20-71-00007, Scaling in sandpile models, 07.2020–06.2022 |
| 2018 | Young Russian Mathematics grant holder, A summation over subsets of $SL(2,\mathbb{Z})$, 01.2018–12.2020 |
| 2016 | PostDoc.Mobility SNSF (Switzerland) grant, P2GEP2_168647, Sandpiles in physics, combinatorics, tropical and algebraic geometry 1.07.2016 – 31.08.2017 |
| 2005 | Gold medal at the International Mathematical Olympiad |

## Service

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|---|---|
| 2020-2024 | Editor-in-chief of a book about St. Petersburg mathematicians, *Saint Petersburg mathematicians and their discoveries*, 496p., (2024), MCCME. Freely available on `https://sites.google.com/view/spbmath`, Russian version is available on `https://biblio.mccme.ru/node/291128` |
| 2020 | Senior Coordinator on International Mathematical Olympiad 2020 |

## Teaching Experience

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|---|---|
| GTIIT: | *Numerical analysis*,<br>*Set theory*,<br>*Algebra B*,<br>*Basic notions of mathematics*,<br>*Infinitesimal analysis 3*<br>*Combinatorial geometry*<br>*Seminar in Geometry*<br>*Probability theory*<br>*Number theory* |
| SPbU: | *Symplectic topology*,<br>*Introduction to homotopy theory*,<br>*Game theory*,<br>*Introduction to homology theory*,<br>*General topology*,<br>*Hyperbolic geometry*,<br>*Topological data analysis*<br>*Differential forms in algebraic topology* |
| HSE: | *Algebra and analysis* (for sociologists),<br>*Game theory* (for economists),<br>*Topological data analysis* (for programmers) |
| UniGe: | *Complex Analysis* (assistant; in French),<br>*Characteristic classes* (assistant; in English),<br>*Méthodes élémentaires* (in French) |

**Master theses under my supervision**

| | |
|---|---|
| 2024 | Triangulations of flat $\mathbb{R}P^2$ with three different conical points |
| 2024 | Sandpile models on oriented graphs |
| 2017 | Sandpile phenomena in tropical systems, Mexico City |
| 2014 | Un polynôme pour les matroïdes signés: généralisations et applications, Geneva |

**Bachelor theses under my supervision**

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|---|---|
| 2024 | Sandpile on infinite graphs |
| 2023 | Newton polytopes of singular surfaces |
| 2023 | Triangulations of flat $\mathbb{R}P^2$ with four conical points |
| 2023 | Densities of lattices of translates |
| 2023 | Patterns in sandpiles |
| 2023 | Growth phenomena in sandpiles |
| 2022 | Analysis of the mechanism of the college admission in Russia in 2021 |
| 2021 | Equilibrium existence in the case of non-divisible objects |
| 2020 | Modeling and analysis of crab auction in Russia |
| 2020 | Repeated auctions with a reserve price |
| 2019 | Isoperimetric problem for a family of closed curves on a sphere |

# Preprints

CayleyPy-4: AI-Holography. Towards analogs of holographic string dualities for AI tasks (joint with other 43 authors), `https://arxiv.org/abs/2603.22195`

Eisenstein-series evaluations for a family of hyperbolic cosine Lambert series, `https://arxiv.org/abs/2603.21265`

Topology of a uniform spanning tree on a cylinder, (joint with D. Rakhmankin) `https://arxiv.org/abs/2602.06383`

A $p$-adic ($p \equiv 3 \pmod 4$) depth-5 supercongruence for Gaussian $p$-th power sums over a square, (joint with F.S. Zottor) `http://arxiv.org/abs/2602.00206`

Weighted error-sum identities for periodic continued fractions and their generalizations, (joint with K. Calderon) `https://arxiv.org/abs/2601.07862`

Limits of equi-affine equi-distant loci of planar convex domains with two non-parallel asymptotes, (joint with M. Shkolnikov) `https://arxiv.org/abs/2512.21154`

A half-shift reflection identity for the digamma function, `https://arxiv.org/abs/2510.00012`

Evaluation of lattice sums via telescoping over topographs, `https://arxiv.org/abs/2510.02082`

Legendre duality for certain summations over the Farey pairs, `https://arxiv.org/abs/2409.10592`

A guide to tropical modifications `https://arxiv.org/abs/1509.03443`

Sandpile group of infinite graphs (joint with V. Khramov), `https://arxiv.org/abs/2305.05346`

Shrinking dynamic on multidimensional tropical series, `https://arxiv.org/abs/2201.07982`

Tropical curves in sandpile models (joint with M. Shkolnikov), `https://arxiv.org/abs/1502.06284`

## Published papers

Evaluating lattice sums via telescoping on $SL_+(2,\mathbb{Z})$: a short proof of $\sum \frac{1}{\|x\|^2\|y\|^2\|x+y\|^2} = \frac{\pi}{4}$ and of Zagier's identity, `https://arxiv.org/abs/2410.10884`

A short telescoping proof of Hata's formula for the Euler–Mascheroni constant, **Integers**, 25, A90 (2025)

Wolstenholme's theorem over Gaussian integers, **Funct. Approx. Comment. Math.** Advance Publication 1-11, (2025)

Tropical Weil's reciprocity law and Weil's pairing (joint with M. Magin, **Journal of Mathematical Sciences**)

(Sandpile patterns on a regular graph of degree eight, in Russian, joint with P. Granin, A. Saakyan) Песочные паттерны на регулярном графе с вершинами степени восемь, **Chebyshevskii Sbornik**, vol. 25 (2024), no. 3, pp.47–69

(Convergence of a sandpile on a triangular lattice under rescaling, in Russian, joint with A. Aliev) Сходимость песочной кучи на треугольной решётке при ремасштабировании, **Matematicheskii Sbornik**, vol. 214 (2023), issue 12, pp.3–25

Some statistics about Tropical Sandpile Model, **Communications in Mathematics**, Volume 31 (2023), Issue 3, 9–19 (joint with Y. Prieto)

Sandpile solitons in higher dimensions, **Arnold Mathematical Journal**, 9 (2023), no. 3, 435–454

Equilateral convex triangulations of $\mathbb{R}P^2$ with three conical points of equal defect, **In the Tradition of Thurston II: Geometry and Groups** (2022, p. 315–329), editors: Ken'ichi Ohshika, Athanase Papadopoulos, Springer (joint with M. Chernaviskikh, A. Erdnigor, and A. Zakharov)

Sandpile Solitons via Smoothing of Superharmonic Functions, **Communications in Mathematical Physics**, 378(3) (2020), 1649–1675 (joint with M. Shkolnikov)

Pattern Formation and Tropical Geometry, **Frontiers in Physics**, 2020 (8), 423

Strategic analysis of the Russian crab quota auction in 2019, **Marine Policy**, Volume 122, December 2020, 104266, (joint with Mark Vershinin)

Tropical formulae for summation over a part of $SL(2,\mathbb{Z})$, **European Journal of Mathematics**, Volume 5 (2019), Issue 3, 909–928 (joint with M. Shkolnikov)

Self-Organized Criticality and Pattern Emergence through the Lens of Tropical Geometry, **PNAS** August 28, 2018. 115 (35) E8135–E8142 (joint with A. Guzmán-Sáenz, Y. Prieto, M. Shkolnikov, V. Kalinina, E. Lupercio)

Legendrian curves in $\mathbb{C}P^3$: cubics and curves on a quadratic surface, **Zapiski Nauchnyh Seminarov PDMI**, vol. 476 (2018), Geometry and Topology Series 13, 92-110 (Journal of Mathematical Sciences,

251(4), 489-502)

Introduction to tropical series and wave dynamic on them, **Discrete and continuous dynamical systems, A**, 38(6) (2018): 2843–2865 (joint with M. Shkolnikov)

The number $\pi$ and summation by $SL(2,\mathbb{Z})$, **Arnold Mathematical Journal**, Volume 3 (2018), Issue 4, 511–517 (joint with M. Shkolnikov)

Tropical approach to Nagata's conjecture in positive characteristic, **Discrete and Computational Geometry**, 58(1) (2017), 158-179

Tropical curves in sandpiles, **Comptes Rendus Mathematique**, Volume 354, Issue 2, 1 February 2016, 125–130 (joint with M. Shkolnikov)

Sandpiles on the heptagonal tiling, **Knot Theory and Its Ramifications**, Vol. 26 (2016), Issue 12, 1642005 (joint with M. Shkolnikov)

The Newton polygon of a planar singular curve and its subdivision, **Journal of Combinatorial Theory, Series A**, 137 (2016), 226–256

The Alexander polynomial as an intersection of two cycles in a symmetric power, **Journal of Knot Theory and Its Ramifications**, Vol. 24 (2015), No. 12, 1550061

**Articles in Russian**

(What could be a dynamical centralized admission system in Russia, in Russian, **Journal of the New Economic Association**) Каким могло бы быть централизованное распределение абитуриентов по образовательным программам, **Журнал Новой Экономической Ассоциации**, № 1 (62), 2024, 101–115 (joint with A. Kuzmina)

(A cipher broken by Ch. Goldbach, in Russian) Шифр, который разгадал Христиан Гольдбах, **Математика в высшем образовании**, 19, с. 85–94, 2021

(Crab auction in 2019: history, estimates and alternative scenarios, in Russian) Крабовый аукцион 2019: история, оценка и альтернативные сценарии, **Экономический журнал ВШЭ**, 2021. Т. 25. № 4. С. 574–594 (joint with D. Ivanov, A. Nesterov, I. Susin)

(Various equilibria in a game-theoretical model of the behavior of a teacher and a student (possibly) cheating on a written exam, in Russian) Различные равновесия в теоретико-игровой модели поведения преподавателя и (возможно) списывающего студента на письменном экзамене, **Социология: методология, методы, математическое моделирование (Социология: 4М)**, вып. 50-51, стр. 97–140

## Recent conference talks

| | |
|---|---|
| 28.03.2026 | "Zeta-series for convex domains", COmbinatorial Number Theory And Connected Topics – V, Ahmedabad University, India |
| 25.09.2025 | "Statistics of Tropical Sandpiles: Degree Growth, Coefficient Laws, and Tree Structure of Curves", Recent Developments on Tropical Sandpiles and Related Subjects, Sofia (ICMS-Sofia), Bulgaria |
| 7.07.2025 | "The residue of a zeta-function for convex domains", The 9th National Conference on Theory of Spaces of Holomorphic Functions, Hulunbuir, China |
| 21.01.2025 | "Zeta-functions for convex domains", MIST: Mathematics Inspired by String Theory 2025, Hong Kong, 21-24 January 2025 |
| 06.01.2025 | "Tropical differential forms on tropical curves", Chinese conference on complex analysis, Shenzhen |

## Recent talks on seminars

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|---|---|
| 19.02.2026 | "Evaluation of lattice sums via telescoping over topographs", Special Functions and Number Theory seminar, India |
| 15.01.2026 | "Sandpile and tropical sandpile models", YMSC Probability seminar, Tsinghua University, Beijing, China |
| 28.11.2025 | "Telescoping over topograph", Colloquium Saint-Peterburg University, Russia |
| 5.11.2025 | "Tropical Weil Reciprocity", Moscow-Beijing Topology Seminar (online) |
| 3.07.2025 | "Zeta function for convex domains", Harbin Polytechnical University, Harbin, China |
| 16.06.2025 | "Sandpile States on a Cylinder and Random Spanning Trees", Jimei University, Xiamen, China |
| 30.04.2025 | "Three Lectures on Binary Quadratic Forms and Conway's Topographs", Sofia, Bulgaria |
| 28.01.2025 | "Elementary SL(2,Z) number theory formulas which stem from the sandpile model", Stavanger, Norway |
| 17.11.2024 | "Number theory in olympiads and scientific life", Harbin Polytechnical University, Harbin, China |
| 20.09.2024 | "Tropical formulae for the number $\pi$", University of Geneva |
| 6.08.2024 | "Several formulae for summation over $SL(2,\mathbb{Z})$", SYSU, Sun Yat-sen University, Guangzhou |
| 21.03.2024 | "Sandpiles on infinite domains", Westlake Math Colloquium, Westlake University, Hangzhou |